benchmark_comparison

March 31, 2026

1 ScyllaDB vs MongoDB — Vector Search Benchmark (2048 Dimensions)

Dataset: 10M vectors · **2048 dimensions** · COSINE similarity (Bioasq 1024-dim extended via dim_override) **Workload:** Search Performance Test (VectorDBBench)

1.0.1 Hardware

	MongoDB	ScyllaDB
Data nodes	3 × M20 (4 GB RAM, 2 CPUs, 20 GB)	3 × i8g.large (16 GB RAM, 2 CPUs, 468 GB)
Search/VS nodes	2 × S50 (32 GB RAM, 4 CPUs)	2 × r7g.xlarge (32 GB RAM, 4 CPUs)
Vectors stored	15.3M	10M
Quantization	scalar	i8 (scalar)

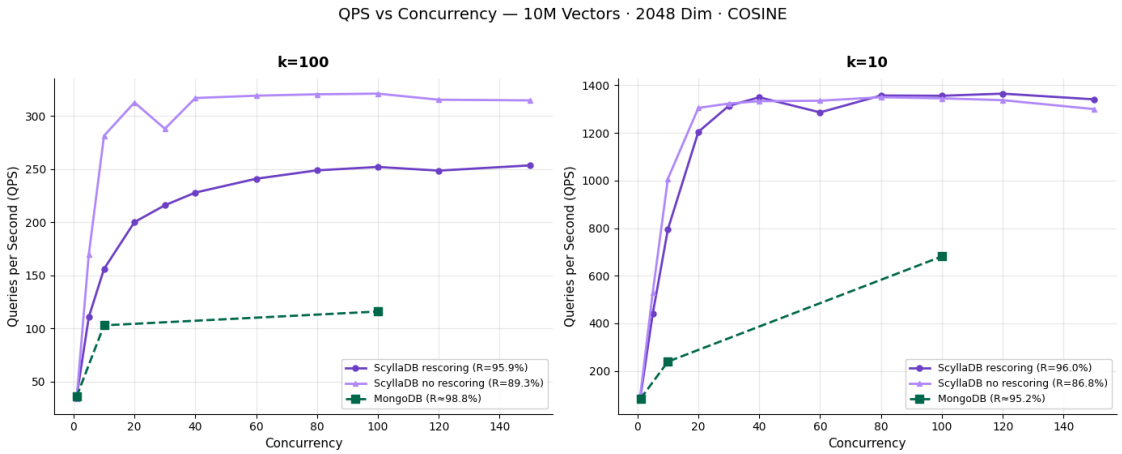
Benchmark client: r7i.16xlarge (64 CPUs), same region as ScyllaDB Cloud (ScyllaDB 2026.1 + VS 1.5.0)

1.0.2 Scenarios

Scenario	Config	k tested
ScyllaDB (rescoring)	oversampling=3.0, rescoring=True, M=64, ef_c=384, ef_s=192	100, 10
ScyllaDB (no rescoring)	oversampling=10.0, rescoring=False, M=64, ef_c=384, ef_s=192	100, 10
MongoDB	Atlas S50 search nodes, scalar quantization	100, 10

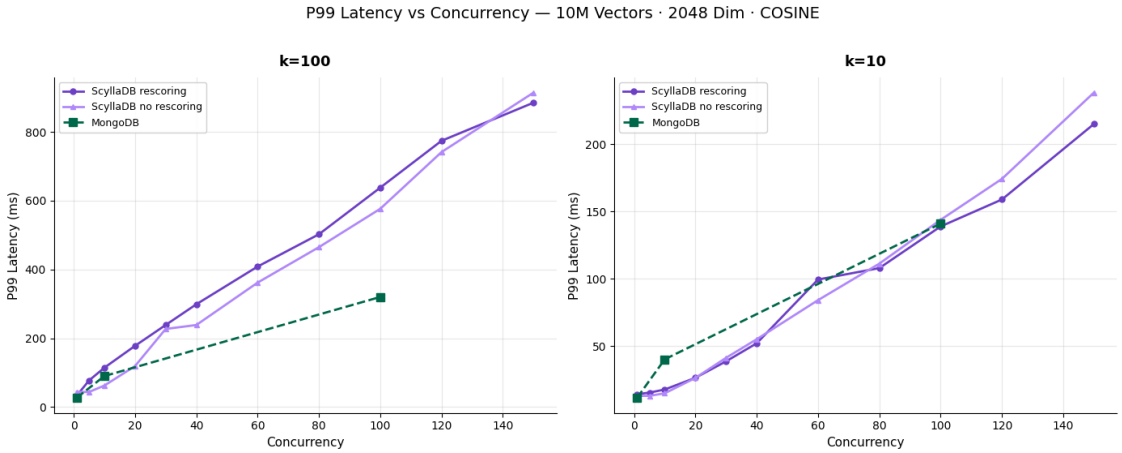
1.1 1. QPS vs Concurrency

How throughput scales with the number of concurrent search clients.

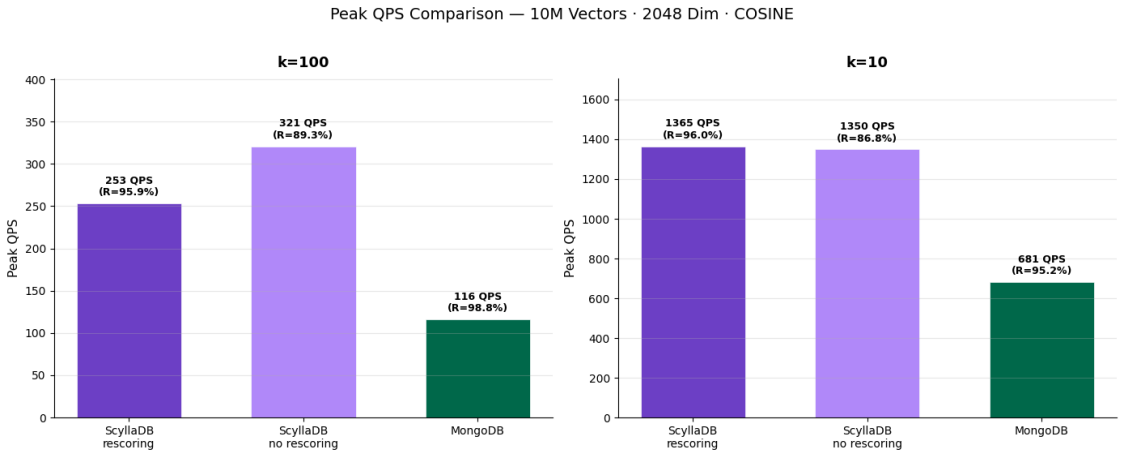


1.2 2. P99 Latency vs Concurrency

Tail latency behavior under increasing load.

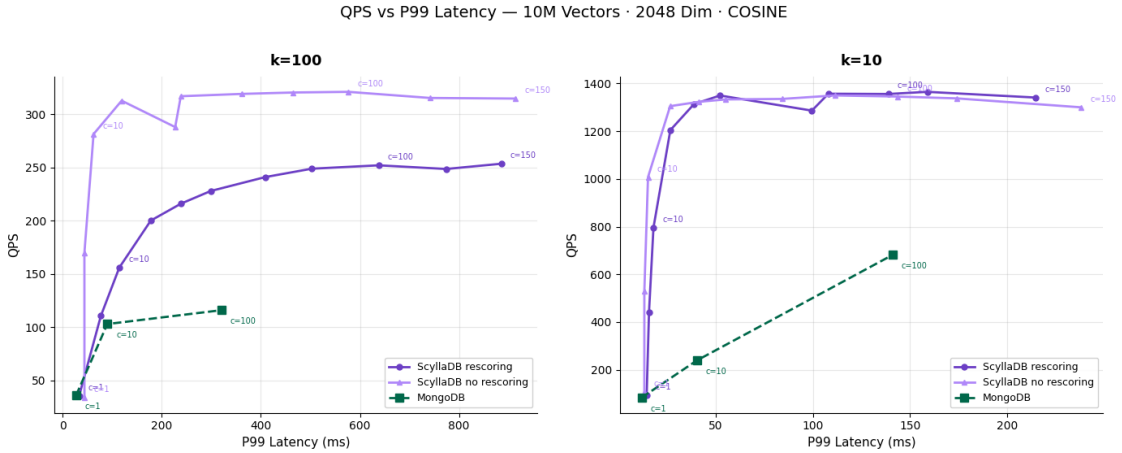


1.3 3. Peak QPS Comparison



1.4 4. QPS vs P99 Latency (Pareto View)

Trade-off between throughput and tail latency. Points closer to the **top-left** corner are better (high QPS, low latency). Concurrency level annotated at key points.



1.5 5. Summary

1.5.1 k=100

Metric	ScyllaDB (rescoring)	ScyllaDB (no rescoring)	MongoDB
Peak QPS	253 (c=150)	321 (c=100)	116 (c=100)
QPS @ c=10	156	281	103
P99 Latency @ c=1	33.2 ms	43.9 ms	27.0 ms
P99 Latency @ c=100	637.8 ms	576.3 ms	320.0 ms

Metric	ScyllaDB (rescoring)	ScyllaDB (no rescoring)	MongoDB
Recall	95.9%	89.3%	~98.8%

1.5.2 k=10

Metric	ScyllaDB (rescoring)	ScyllaDB (no rescoring)	MongoDB
Peak QPS	1,365 (c=120)	1,350 (c=80)	681 (c=100)
QPS @ c=10	796	1,008	239
P99 Latency @ c=1	14.4 ms	13.2 ms	11.8 ms
P99 Latency @ c=100	138.9 ms	143.8 ms	141.0 ms
Recall	96.0%	86.8%	~95.2%

1.5.3 Key observations

- **k=100:** ScyllaDB with rescoring achieves **2.2× higher peak QPS** than MongoDB (253 vs 116) at comparable recall (~96%); no-rescoring pushes to **2.8×** (321 vs 116) but recall drops to 89.3%
- **k=10:** ScyllaDB dominates — **2.0× peak QPS** with rescoring (1,365 vs 681) at similar recall; no-rescoring is comparable at 1,350 QPS but lower recall (86.8%)
- **Rescoring trade-off:** Rescoring yields ~7% higher recall at k=100 with ~21% lower peak QPS; at k=10 the QPS difference is negligible
- **Latency:** At low concurrency (c=1), all three scenarios show comparable P99 latencies (11–44 ms); MongoDB maintains lower tail latency at high concurrency for k=100
- **Scaling:** ScyllaDB throughput scales smoothly up to c=150; MongoDB was tested up to c=100